Five Integrals Converging to π

A Pi Day Collection

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Editorial Note

Today we celebrate **Pi Day**, the annual homage to the most celebrated constant in all of mathematics. March 14 $(3/14)$ mirrors the opening digits of $\pi = 3.14\ldots$, a number that quietly governs geometry, probability, physics, and analysis alike.

In the spirit of this occasion, the **Namssn Academic Team** presents five integrals — ranging from the classical to the extraordinary — each of which, through distinct and elegant techniques, resolves to an expression involving π. These are not trivial computations; they are among the most beautiful results in mathematical analysis. Each solution is presented in full, step-by-step, with the intent of both instruction and appreciation.

Happy Pi Day. May the circle never be squared.

— Abdulrasheed, Editor

The Five Integrals at a Glance:

#	Integral	Result
I	$\displaystyle\int_{-\infty}^{\infty} e^{-x^2}\,dx$	$\sqrt{\pi}$
II	$\displaystyle\int_{0}^{\infty} \frac{\sin x}{x}\,dx$	$\dfrac{\pi}{2}$
III	$\displaystyle\int_{0}^{1} \frac{x^4(1-x)^4}{1+x^2}\,dx$	$\dfrac{22}{7}-\pi$
IV	$\displaystyle\int_{0}^{\pi/2} \ln(\sin x)\,dx$	$-\dfrac{\pi}{2}\ln 2$
V	$\displaystyle\int_{0}^{1} \frac{\arctan\left(\sqrt{x^2+2}\right)}{(1+x^2)\sqrt{x^2+2}}\,dx$	$\dfrac{5\pi^2}{96}$

1. The Gaussian Integral

$$I = \int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$$

Strategy

The integrand e^{-x^2} admits no elementary antiderivative. The key insight is to consider I^2, convert the resulting double integral to polar coordinates, and evaluate directly.

Step 1: Square the integral and write as a double integral.

Let $I = \int_{-\infty}^{\infty} e^{-x^2}\,dx$. Since the dummy variable may be renamed:

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2}\,dx\right)\left(\int_{-\infty}^{\infty} e^{-y^2}\,dy\right) = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)}\,dx\,dy.$$

Step 2: Switch to polar coordinates.

Set $x = r\cos\theta$, $y = r\sin\theta$, with Jacobian $dA = r\,dr\,d\theta$, and $x^2+y^2 = r^2$:

$$I^2 = \int_0^{2\pi}\int_0^{\infty} r e^{-r^2}\,dr\,d\theta.$$

Step 3: Evaluate the inner radial integral.

Substitute $u = r^2$, $du = 2r\,dr$:

$$\int_0^{\infty} r e^{-r^2}\,dr = \frac{1}{2}\int_0^{\infty} e^{-u}\,du = \frac{1}{2}\Big[-e^{-u}\Big]_0^{\infty} = \frac{1}{2}.$$

Step 4: Evaluate the angular integral.

$$I^2 = \int_0^{2\pi} \frac{1}{2}\,d\theta = \frac{1}{2}\cdot 2\pi = \pi.$$

Step 5: Conclude.

Since $I > 0$:

$$\boxed{\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}.}$$

2. The Dirichlet Sinc Integral

$$I = \int_0^\infty \frac{\sin x}{x}\,dx = \frac{\pi}{2}$$

Strategy

The integrand is not absolutely integrable over $[0,\infty)$, so standard techniques fail. We employ **Feynman's trick** — differentiation under the integral sign — by introducing a decay parameter t and solving the resulting ODE.

Step 1: Define the parametric family.

Set

$$I(t) = \int_0^\infty \frac{\sin x}{x} e^{-tx}\,dx, \quad t \geq 0.$$

Note $I(\infty) = 0$ and our target is $I(0)$.

Step 2: Differentiate under the integral sign.

$$I'(t) = \frac{d}{dt}\int_0^\infty \frac{\sin x}{x} e^{-tx}\,dx = -\int_0^\infty e^{-tx}\sin x\,dx.$$

Step 3: Evaluate the integral for $I'(t)$.

Recognise $\sin x = \mathrm{Im}(e^{ix})$:

$$\int_0^\infty e^{-tx}\sin x\,dx = \mathrm{Im}\left[\int_0^\infty e^{-(t-i)x}\,dx\right] = \mathrm{Im}\left[\frac{1}{t-i}\right] = \mathrm{Im}\left[\frac{t+i}{t^2+1}\right] = \frac{1}{t^2+1}.$$

Hence:

$$I'(t) = -\frac{1}{1+t^2}.$$

Step 4: Integrate from t to ∞.

Using $I(\infty) = 0$:

$$0 - I(t) = -\int_t^\infty \frac{dt'}{1+t'^2} = -\left(\frac{\pi}{2} - \arctan t\right),$$

$$\therefore \quad I(t) = \frac{\pi}{2} - \arctan t.$$

Step 5: Set $t = 0$.

$$\boxed{\int_0^\infty \frac{\sin x}{x}\,dx = I(0) = \frac{\pi}{2}.}$$

3. The 22/7 − π Integral

$$I = \int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,dx = \frac{22}{7} - \pi$$

Strategy

Expand the numerator via the binomial theorem, perform polynomial long division by $1+x^2$, then integrate term by term. The arctangent integral yields π.

Step 1: Expand the numerator.

$$x^4(1-x)^4 = x^4\left(1-4x+6x^2-4x^3+x^4\right) = x^8-4x^7+6x^6-4x^5+x^4.$$

Step 2: Polynomial long division of $x^8-4x^7+6x^6-4x^5+x^4$ by $1+x^2$.

Step	Quotient Term	Subtracted	Remainder
1	x^6	x^8+x^6	$-4x^7+5x^6-4x^5+x^4$
2	$-4x^5$	$-4x^7-4x^5$	$5x^6+x^4$
3	$5x^4$	$5x^6+5x^4$	$-4x^4$
4	$-4x^2$	$-4x^4-4x^2$	$4x^2$
5	4	$4x^2+4$	-4

Therefore:

$$\frac{x^4(1-x)^4}{1+x^2} = x^6-4x^5+5x^4-4x^2+4-\frac{4}{1+x^2}.$$

Step 3: Integrate term by term from 0 to 1.

$$I = \left[\frac{x^7}{7}-\frac{4x^6}{6}+\frac{5x^5}{5}-\frac{4x^3}{3}+4x-4\arctan x\right]_0^1 = \frac{1}{7}-\frac{2}{3}+1-\frac{4}{3}+4-4\cdot\frac{\pi}{4}.$$

Step 4: Collect rational and transcendental parts.

$$\text{Rational part:}\quad \frac{1}{7}-\frac{2}{3}+1-\frac{4}{3}+4 = \frac{1}{7}+5-2 = \frac{1}{7}+3 = \frac{22}{7}.$$

$$\boxed{\int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,dx = \frac{22}{7} - \pi.}$$

Remark. Since the integrand is strictly positive on $(0,1)$, the result $I > 0$ constitutes a rigorous proof that $\dfrac{22}{7} > \pi$.

4. Euler's Log-Sine Integral

$$I = \int_0^{\pi/2} \ln(\sin x)\,dx = -\frac{\pi}{2}\ln 2$$

Strategy

We exploit the symmetry between sin and cos over $[0, \pi/2]$, reduce to a self-referential equation using the double-angle identity, and solve algebraically.

Step 1: Observe the cosine counterpart.

Substitute $x \mapsto \frac{\pi}{2} - x$:

$$I = \int_0^{\pi/2} \ln(\cos x)\,dx.$$

Step 2: Add the two expressions for I.

$$2I = \int_0^{\pi/2} \big[\ln(\sin x) + \ln(\cos x)\big]\,dx = \int_0^{\pi/2} \ln(\sin x \cos x)\,dx.$$

Step 3: Apply the double-angle identity.

Using $\sin x \cos x = \frac{1}{2}\sin 2x$:

$$2I = \int_0^{\pi/2} \ln\big(\tfrac{1}{2}\sin 2x\big)\,dx = \int_0^{\pi/2} \ln(\sin 2x)\,dx - \frac{\pi}{2}\ln 2.$$

Step 4: Transform the remaining integral.

Let $u = 2x$:

$$\int_0^{\pi/2} \ln(\sin 2x)\,dx = \frac{1}{2}\int_0^{\pi} \ln(\sin u)\,du.$$

By the symmetry $\sin(\pi - u) = \sin u$:

$$\int_0^{\pi} \ln(\sin u)\,du = 2\int_0^{\pi/2} \ln(\sin u)\,du = 2I.$$

Step 5: Solve the functional equation.

$$2I = \frac{1}{2}\cdot 2I - \frac{\pi}{2}\ln 2 = I - \frac{\pi}{2}\ln 2,$$

$$I = -\frac{\pi}{2}\ln 2.$$

$$\int_0^{\pi/2} \ln(\sin x)\,dx = -\frac{\pi}{2}\ln 2.$$

5. The Ahmed Integral

$$I=\int_0^1 \frac{\arctan\left(\sqrt{x^2+2}\right)}{(1+x^2)\sqrt{x^2+2}}\,dx=\frac{5\pi^2}{96}$$

Strategy

This integral, published by Z. Ahmed in 2002, requires three layers: (1) a complementary arctangent identity splits I into two parts; (2) part P is evaluated via a hyperbolic substitution; (3) part L is handled by Feynman's trick, which produces a self-referential equation.

Part A: Evaluating P

Step 1: Complementary arctangent split.

Use $\arctan\theta+\arctan\frac{1}{\theta}=\frac{\pi}{2}$ for $\theta>0$:

$$\arctan\left(\sqrt{x^2+2}\right)=\frac{\pi}{2}-\arctan\frac{1}{\sqrt{x^2+2}}.$$

Decompose:

$$I=\underbrace{\frac{\pi}{2}\int_0^1 \frac{dx}{(1+x^2)\sqrt{x^2+2}}}_{P}-\underbrace{\int_0^1 \frac{\arctan\frac{1}{\sqrt{x^2+2}}}{(1+x^2)\sqrt{x^2+2}}\,dx}_{L}.$$

Step 2: Hyperbolic substitution for P.

Let $x=\sqrt{2}\sinh u$, so $dx=\sqrt{2}\cosh u\,du$ and $\sqrt{x^2+2}=\sqrt{2}\cosh u$:

$$\int_0^1 \frac{dx}{(1+x^2)\sqrt{x^2+2}}=\int_0^{u_1}\frac{du}{1+2\sinh^2 u}=\int_0^{u_1}\operatorname{sech}(2u)\,du=\frac{1}{2}\Big[\arctan(\sinh 2u)\Big]_0^{u_1}.$$

At the upper limit $u_1=\operatorname{arcsinh}(1/\sqrt{2})$: $\sinh u_1=\frac{1}{\sqrt{2}}$, $\cosh u_1=\sqrt{\frac{3}{2}}$, so $\sinh 2u_1=\sqrt{3}$. Therefore:

$$\int_0^1 \frac{dx}{(1+x^2)\sqrt{x^2+2}}=\frac{1}{2}\arctan(\sqrt{3})=\frac{\pi}{6},\qquad P=\frac{\pi}{2}\cdot\frac{\pi}{6}=\frac{\pi^2}{12}.$$

Part B: Evaluating L via Feynman's Trick

Step 3: Introduce the parameter integral.

$$L(t)=\int_0^1 \frac{\arctan\frac{t}{\sqrt{x^2+2}}}{(1+x^2)\sqrt{x^2+2}}\,dx,\qquad L(0)=0,\quad L(1)=L.$$

Step 4: Differentiate under the integral sign.

$$L'(t) = \int_0^1 \frac{dx}{(1+x^2)(x^2+2+t^2)}.$$

Partial fractions:

$$\frac{1}{(1+x^2)(x^2+2+t^2)} = \frac{1}{1+t^2}\left(\frac{1}{1+x^2} - \frac{1}{x^2+2+t^2}\right).$$

Hence:

$$L'(t) = \frac{1}{1+t^2}\left(\frac{\pi}{4} - \frac{\arctan\frac{1}{\sqrt{t^2+2}}}{\sqrt{t^2+2}}\right).$$

Step 5: Integrate $L'(t)$ from 0 to 1.

$$L = \frac{\pi}{4}\int_0^1 \frac{dt}{1+t^2} - \int_0^1 \frac{\arctan\frac{1}{\sqrt{t^2+2}}}{(1+t^2)\sqrt{t^2+2}}\,dt = \frac{\pi^2}{16} - L.$$

The second integral is exactly L with t as dummy variable.

Step 6: Solve the self-referential equation.

$$2L = \frac{\pi^2}{16} \implies L = \frac{\pi^2}{32}.$$

Final Assembly

$$I = P - L = \frac{\pi^2}{12} - \frac{\pi^2}{32} = \pi^2\left(\frac{8}{96} - \frac{3}{96}\right).$$

$$\boxed{\int_0^1 \frac{\arctan\left(\sqrt{x^2+2}\right)}{(1+x^2)\sqrt{x^2+2}}\,dx = \frac{5\pi^2}{96}.}$$

Remark. The decisive moment is Step 3 of Part B: Feynman's trick generates a mirror of L on the right-hand side, allowing it to be solved without ever computing it directly.

"The mathematician's patterns, like the painter's or the poet's, must be beautiful."
— G.H. Hardy

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